

Detailed Report: Dataset Profiling (Milestone-1)

This report provides a technical breakdown of the exploratory data analysis and initial audio processing performed in Milestone 1.

1. File Integrity and Corruption Analysis

The first step in the pipeline was to audit the raw dataset for physical errors or incomplete downloads. I implemented a scanning script to categorize files based on their size:

- Corruption Check: A threshold of 4KB was set to identify corrupted or empty .wav headers. The scan confirmed 0 corrupted files within the directory.
- Size Variance: I analyzed files against a boundary of ~5.05MB.
 - Wav files < 5.0491MB: 1,256 files.
- Significance: The absolute difference of 1256
- files suggests that while the majority of the dataset follows a standard duration/bitrate, there is a significant subset of files with higher bitrates or longer durations that may require normalization during training.

2. Dataset Construction: build_dataset

The build_dataset function is the core logic for transforming a flat folder structure into a structured, reproducible dataset.

Core Logic:

- Genre-Stem Grouping: Instead of treating audio files as individual samples, the function groups four specific stems - drums, vocals, bass, and other - into a single "song". This ensures that when we train a model, all components of a single song are kept together.
- Validation Logic: For every song folder, the function checks for the existence of all four required .wav files and verifies they exceed the 4KB corruption limit.
- Deterministic Splitting: Using a val_split of 0.17 and a fixed seed, the function shuffles the "song" and divides them.
- Outcome: This resulted in a balanced split where each of the 10 genres contains 83 training songs and 17 validation songs. For example, the absolute difference between training reggae drums and validation country vocals is 66 (83 - 17).

3. Silence Profiling: find_long_silences

Because this dataset contains "stems" (isolated instruments), many tracks contain long periods where the instrument is not playing. The find_long_silences function identifies these regions to prevent models from learning to "generate" empty air.

Implementation Steps:

1. Loading: Audio is loaded at SR = 22,050 Hz.
2. Splitting: It uses librosa.effects.split with a top_db=20 threshold. Any audio below this decibel level is considered "silence".
3. Interval Calculation: The function calculates the time gaps between non-silent intervals. If a gap (at the start, middle, or end) exceeds 5 seconds, it is logged.
4. Metadata Tagging: It records the location (Start, Middle, End) and the maximum continuous duration of the silence.

4. Data Analysis

The analysis following the silence detection reveals critical insights into the "Messy Mashup" dataset's characteristics.

A. The "Vocal" Irregularity

The analysis showed that Vocals are the most inconsistent stem.

- 304 vocal files contain silences longer than 5 seconds.
- The average silence length in these vocal tracks is 12.59s, indicating that singers are often absent for nearly half of a standard 30-second clip.

B. Genre-Specific Findings (Jazz Drums)

We performed a deep dive into the Jazz genre's percussion:

- 24 Jazz drum tracks have silences > 5s.
- 13 tracks have silence located strictly in the middle, suggesting "breaks" or solo sections for other instruments.
- 7 tracks feature extreme silence gaps of 10 seconds or more.

C. Pivot Table Analysis (Genre vs. Stem)

A distribution analysis of silent files across the dataset:

--- Pivot Table (Genre vs Stem) ---				
Stem	bass	drums	other	vocals
Genre				
blues	17	22	5	43
classical	68	57	5	69
country	16	16	2	16
disco	8	2	3	18
hiphop	20	3	22	6
jazz	25	24	1	70
metal	6	2	1	40
pop	10	5	2	4
reggae	5	4	7	12
rock	10	7	1	26

Observation: Classical and Jazz show the highest rates of stem silence, likely due to the dynamic arrangements inherent in those genres, whereas Pop and Disco have more continuous instrumental presence.

5. Mixing and Normalization Baseline

Finally, a baseline mix was generated for Rock Song Index 0 to test the summation math:

- Mixing: Stems were stacked and summed.
- RMS Amplitude: The raw mixture's RMS was 0.1021, indicating the average "loudness" before processing.
- Peak Normalization: To prevent digital distortion (clipping), the mix was scaled so the maximum absolute value is exactly 1.0.
- Sample Count: The 5-second mix consists of exactly 110,250 samples.